

Student Scientific Misconduct

As a Grand Award Judge, you may come across behavior by a student that you regard as suspect. This document clarifies how you should (and should not) respond.

Improper Student Behavior

Similar to professional scientific and engineering researchers, ISEF finalists are expected to honestly and accurately perform and report their work. As a condition of acceptance into ISEF, all finalists agree to the ethics statement included in the appendix to this document, the summary paragraph of which is as follows:

Scientific fraud and misconduct are not condoned at any level of research or competition. This includes plagiarism, forgery, use or presentation of other researcher's work as one's own and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and Regeneron ISEF. The Society for Science reserves the right to revoke recognition of a project subsequently found to have been fraudulent.

Detecting Scientific Misconduct

As in professional research, scientific misconduct among student projects at ISEF is rare. However, if you believe that you have uncovered such misconduct, you should attempt to confirm your suspicions before taking any action on them. You have multiple opportunities to familiarize yourself with the content of every project in your category:

1. About ten days before the fair, judges will be able to view complete information about every project. This information includes a short abstract, a one-page *Quad Chart* summarizing the project, a *Project Presentation* that can be up to 12 pages long, and a short video introduction. Some projects may also provide a research paper, though this is not a required document.
2. On Tuesday, judges have all afternoon (following Registration) and all evening (until 9:00 pm) to examine each project display.
3. On Wednesday morning before students arrive, Grand Award judges have until 7:40 am again to examine each project display.

Plagiarism, or presenting work by others as the student's own, should be detectable from a review by subject experts of the project's materials available online. In some cases, it may take a review of the project display itself on Tuesday afternoon or evening to trigger a recollection of prior work. In all cases, judges have more than sufficient time before judging begins on Wednesday to do the necessary background research before lodging a concern about scientific misconduct.

Reporting Suspected Misconduct

If you believe that you have confirmed your suspicions of misconduct, do not confront the student(s).¹ Instead, present your concern, along with any relevant evidence, to your category co-chairs who will bring your case to core Fair personnel for further investigation and determination. Given the plentiful opportunities for early detection, **ISEF requires all claims of scientific misconduct to be reported as soon as the concerns are uncovered and in no case any later than the beginning of Interview Period 1 at 9:30 am.**

Results of the Fair's Investigation

Core Fair personnel will investigate every report of possible student misconduct using all resources including contacting the original reporter as well as other independent subject experts. The outcome of that investigation will be communicated to the project's category co-chairs as soon as possible, and in all cases no later than the end of Interview Period 20.

- If the concern is found to be unwarranted, then other than co-chairs reporting that result back to you, no further action will be taken. Since no one else was notified of the investigation, no one else needs to know that it has concluded.
- If, however, the Fair substantiates your concern, then your co-chairs, and only your co-chairs, will have the responsibility of reporting the result of that investigation during the caucusing. The Fair's investigation may determine that there is clear misconduct, but it may also decide that there is sufficient reason to be suspicious, but is not clear-cut and should be discussed in caucus by all judges.

If the project is never considered for any award,² the co-chairs will never mention the investigation as it would be irrelevant to caucus concerns. Only if the project is considered for an award² will the co-chairs inform the caucus of the investigation and the Fair's determination.

Prohibited Judge Conduct

Because the responsibility for determination of scientific misconduct lies with the Fair, and the responsibility for reporting that determination lies with the co-chairs, **judges may not themselves introduce their suspicions of scientific misconduct during the caucus.**

Raising this concern in the caucus, that is, at the last minute, would unfairly saddle a project with the suspicion of impropriety without the opportunity for independent investigation of these claims. If any judge raises such a claim during the caucusing, the co-chairs will disallow it, and all judges should ignore the claim in their deliberations.

¹ If you do interview the student(s), whether as a scheduled or unscheduled interview, you may ask the student factual questions about your concern, but do not do so in an accusatory manner. Just as importantly, do not reveal that you have reported your concern to the Fair.

² "Considered for an award" means that the project is discussed for possible addition to any award level's Pool of Award Candidates, well before any voting on an actual award. See Section V of the Judge Information Board concerning *The ISEF Award Decision Process* for further details.

Points of Emphasis

1. If you think a student has engaged in scientific misconduct of any nature, you must bring your concern with evidence to one of your category co-chairs as soon as possible, and in no case any later than the start of Interview Period 1 at 9:30 am on Wednesday.
2. Category co-chairs will report to the caucus at the appropriate time any confirmed ethical violations plus any that the Fair could not substantiate but deems worthy of caucus consideration.
3. During caucusing for awards, judges may consider the issue of student scientific misconduct for a given project only if the category co-chairs have introduced it. ISEF prohibits judges from raising this issue on their own during the caucus.

Appendix: Ethics Statement for Students

Source: <https://www.societyforscience.org/isef/international-rules/rules-for-all-projects/>

All student participants must agree to the following statement of expectations as a condition of acceptance into the Regeneron International Science and Engineering Fair.

Ethics Statement

Student researchers, as well as adults who have a role in their projects, are expected to maintain the highest ethical standards. These include, but are not limited to:

- **Integrity.** Honesty, objectivity, and avoidance of conflicts of interest are expected during every phase of the research. The project should reflect independent research done by the student(s), and represent only one year's work.
- **Legality.** Compliance with all federal, state and local laws and regulations is essential. In addition, projects conducted outside the U.S. must also adhere to the laws of the country and jurisdiction in which the project was performed. All projects must be approved by a Scientific Review Committee (SRC), and when necessary must also be approved by an Institutional Review Board (IRB), Institutional Animal Care and Use Committee (IACUC), and/or Institutional Biosafety Committee (IBC).
- **Respect for Confidentiality and Intellectual Property.** Confidential communications, as well as patents, copyrights, and other forms of intellectual property must be honored. Unpublished data, methods, or results may not be used without permission, and credit must be given to all contributions to research.
- **Stewardship of the Environment.** It is the responsibility of the researcher and the adults involved to protect the environment from harm. Introduction or disposal of non-native, genetically-altered, and/or invasive species, (e.g. insects, plants, invertebrates, vertebrates), pathogens, toxic chemicals or foreign substances into the environment is prohibited. It is recommended that students reference their local, state or national laws and regulations and quarantine lists, including if considering using "catch and release" fishing procedures.
- **Animal Care.** Proper care and respect must be given to vertebrate animals. The guiding principles for the use of animals in research includes the following "Four R's": Replace, Reduce, Refine, Respect.
- **Human Participant Protection.** The highest priority is the health and well-being of the student researcher(s) and human participants.
- **Potentially Hazardous Biological Agents (PHBAs).** It is the responsibility of the student and adults involved in the project to conduct and document a risk assessment, and to safely handle and dispose of organisms and materials.

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